

Paying Attention to the Dips: Transformer-based Exoplanet Detection

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1 Abstract

Exoplanets, planets that orbit stars outside our solar system, pose an intriguing prospect: if their conditions are right, they can potentially harbor life. Examining light curves from NASA’s *Kepler* and *Transiting Exoplanet Survey Satellite* (TESS) can help us identify exoplanets; however, data from these sources are noisy, exoplanet detection is very rare, and manual evaluation of light curves is labor-intensive. To address this problem, deep learning methods, like CNNs, have been applied, but transformer-based approaches are largely underexplored. In this project, we compare a multi-input transformer that incorporates local and global views of light curves, to the performance of a CNN baseline. We also look at the effects of data augmentation and self-supervised learning (SSL) on Transformer performance. Our CNN achieved an Area Under Precision Recall Curve (PR-AUC) of 0.83, and the Transformer variants’ PR-AUC ranged from 0.81-0.83. The Transformers did not outperform our CNN on the precision-recall trade-off, but they did achieve much higher recall (0.85-0.86 vs 0.77), indicating greater sensitivity to uncommon signals of exoplanet presence and improved performance for novel exoplanet detection, given the scarcity of confirmed exoplanets in the TESS dataset. The high recall, especially for our transformer trained with SSL, indicates the potential to identify exoplanets using unseen data; however, transformer models have limitations, and future directions could include using sparse-attention mechanisms with unfolded light curves and larger datasets to improve classification performance.

Github Repository: <https://github.com/savula13/Exoplanet-Detection>

2 Introduction

The search for exoplanets, or planets orbiting stars beyond our solar system, relies heavily on the detection of subtle, periodic dips in stellar brightness as planets transit their host stars. Missions such as NASA’s *Kepler* and *Transiting Exoplanet Survey Satellite* (TESS) have deployed instruments that

collect light intensity (flux) information over time. Numerous factors can contribute to dips in the flux measurements over time: binary star systems; instrument noise; sun spots; and the transit of exoplanets in front of our view of stars [8]. This has produced massive quantities of such brightness time series, or light curves.

It is challenging to identify the existence of exoplanets by observing a dip in light on a light time series graph. Previously, experts considered all available data for a given candidate exoplanet light curve to manually determine whether the signal represented a true planetary transit or a false positive [7].

Recently, deep learning techniques have shown success in this classification task. Convolutional Neural Networks (CNNs) and transformer architectures have shown significant promise [6, 7].

Despite these advances in applying deep learning, the field continues to face significant challenges: the scarcity of large, labeled datasets; extreme class imbalance between confirmed exoplanets and non-planetary signals; and the high computational cost of training transformers on long, high-resolution sequences. With the recent construction of the TESS dataset, we aim to evaluate and refine transformer-based methods for exoplanet detection from TESS light curves, and to compare their performance against CNN baselines.

This project aims to determine whether transformers can achieve improved precision and specificity, two important metrics in a field where false positives are expensive, while maintaining recall performance.

3 Related Work & Background

Early data mining approaches to exoplanet detection relied on domain expertise and statistical filtering. For example, machine learning classifiers such as decision trees and Random Forests were used to distinguish planetary transits from false positives, but these methods required carefully engineered features and often underperformed on noisy data [8].

The introduction of deep learning marked an important turning point. Shallue and Vanderburg demonstrated that

convolutional neural networks could automatically identify exoplanet candidates from Kepler’s light curves, surpassing classical techniques in recall [8]. CNNs use local convolutional filters to detect short-term dips in brightness corresponding to potential transits. However, they struggle to represent long-term temporal dependencies [7]. This is an important limitation, given that transit events occur over periodic intervals that may span hundreds or thousands of time steps.

ExoMiner and Exominer++ are two more applications of deep learning for exoplanet classification [9, 10]. ExoMiner identified new planets from the MAST Kepler Archive, and focused on explainability and generalization [10]. ExoMiner++ improves transit signal classification in TESS data, and adds more types of inputs for flux and stellar, improving recall and robustness to noise[9].

Subsequent research expanded the scope of deep learning in astrophysics. Cuéller et al. used synthetic and real light curves jointly for data augmentation, mitigating dataset scarcity and improving generalization [2]. Salinas et al. explored transformer-based architectures for planetary transit classification, demonstrating improved robustness to noise and the ability to model long-range temporal relationships [6]. Similar methods have also been applied in other sequential domains such as ECG signal analysis, where hybrid CNN-Transformer architectures achieved superior performance in identifying complex patterns [3].

Given the rarity of exoplanet detection and the imbalances in the TESS dataset, self-supervised learning (SSL) is an appropriate method to leverage unlabeled data to improve model performance. Li et al. trained a classifier to distinguish between normal and augmented images to learn representations and identify anomalous defects [4]. Luo et. al used psuedo labeling to improve prediction performance on a storngly class imbalanced image dataset with confidence thresholds [5]. Yaman et al. demonstrated the effectiveness of using physics-guided reconstruction with masking for self-supervised learning for MRI images [11].

These works collectively suggest that attention-based models, potentially combined with convolutional front-ends and self-supervised learning approaches, could be well-suited for detecting weak, temporally distributed signals such as exoplanet transits. However, few studies have directly compared transformers to CNNs using TESS data.

4 Proposed Method

4.1 Data Description

Each light curve represents a time series of stellar flux values over time. Before training, light curves undergo detrending to remove long-term brightness variation and normalization to ensure consistent scale across samples.

The flux values are phase-folded aligning periodic transit events based on estimated orbital periods. This folding

creates an average representation of a transit event to better reveal the true structure of the transit as seen through flux values while smoothing out noise.

Each processed light curve is represented as a fixed-length sequence

$$X = [x_1, x_2, \dots, x_T]$$

where x_t denotes normalized flux at time step t , and T is the total number of sampled timesteps.

To better capture event locality, each phase-folded light curve is provided in two complimentary views:

- Global view - phase-folded flux across the orbital period
- Local view - a magnified window centered on the region of greatest flux change, highlighting the potential transit event

In addition to flux measurements, the dataset also includes a centroid position (also referred to as the transit peak location). This value identifies the phase index where the transit is most likely to occur. It provides a feature that can be used to enhance localization, guide the formation of the local view, or support downstream attention mechanisms in our transformer-based models.

Figure 1 shows the two-view representation for a confirmed planet candidate. The Figure plots the global view on the left, which is the phase-folded flux over the entire phase, where each view is a 1-D tensor of size 301. On the right is the local view, which is the phase-folded flux over a subset of the phase with the greatest change in flux, where each view is a 1-D tensor of size 31.

Each light curve also has an identified centroid, or

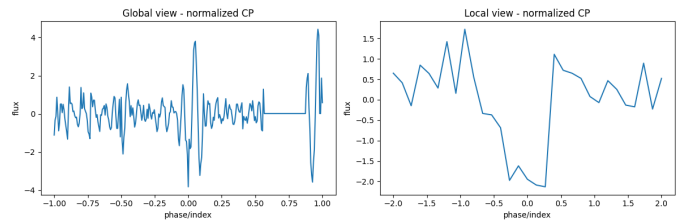


Figure 1. Global and Local Flux Views of Confirmed Planet

4.2 CNN Architecture & Training

Our baseline classifier is a 1D convolutional neural network designed to capture local temporal structure in phase-folded light curves. The architecture follows the successful approach of Shalle and Vanderburg [8], while adapting to the TESS two-view flux representation.

In this model we use only the global flux view as input, represented as a fixed-length 301 element sequence¹. Each

¹The length 301 corresponds to the global view resolution described in Section 3.1. After preprocessing, we observed a post-tensor-conversion length of 296 in the HDF5 dataset. The final network parameters are sized according to the actual 296-step input.

light curve is treated as a single-channel time series, and no centroid features are explicitly provided to the network at this stage.

The CNN consists of three convolutional blocks, each composed of a 1D convolution, batch normalization, ReLU activation, and max-pooling. These operations progressively downsample the sequence ($296 \rightarrow 148 \rightarrow 74 \rightarrow 37$), increasing the number of learned filters ($32 \rightarrow 64 \rightarrow 128$) to extract increasingly abstract transit features. The resulting feature map is flattened and passed through a fully connected projection of size $4736 \rightarrow 256$ with dropout ($p = 0.5$) before the final binary classification layer outputs logits for $\{planet, nonplanet\}$.

The full architecture is shown in Figure 2 and is described as:

$$\begin{aligned}
 &Conv1d(1 \rightarrow 32, k = 7) \rightarrow BN \rightarrow ReLU \rightarrow MaxPool(2) \\
 &Conv1d(32 \rightarrow 64, k = 5) \rightarrow BN \rightarrow ReLU \rightarrow MaxPool(2) \\
 &Conv1d(64 \rightarrow 128, k = 3) \rightarrow BN \rightarrow ReLU \rightarrow MaxPool(2) \\
 &Flatten \rightarrow Linear(4736 \rightarrow 256) \rightarrow Dropout(0.5) \\
 &Linear(256 \rightarrow 2)
 \end{aligned} \tag{1}$$

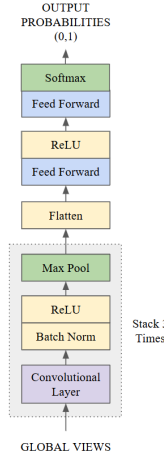


Figure 2. CNN architecture

We optimize the model using Adam (learning rate 1×10^{-3}). Due to the class imbalance, false positives vastly outnumber confirmed planets, so we replace standard binary cross-entropy with Focal Loss ($\alpha = 0.25$, $\gamma = 2$), which down-weights well-classified non-planet examples and increases sensitivity to rare planetary transit patterns. Training is performed with stratified splits of the labeled TESS dataset (75% train/15% validation/15% test) to preserve class proportion.

This model serves as a performance baseline and as a means to confirm that local temporal patterns alone provide a strong discriminative signal for exoplanet classification

before introducing global-context attention mechanisms in later architectures.

4.3 Transformer Architecture

To capture long-range dependencies that CNNs struggle with, we introduced a self-attention-based architecture designed specifically for multi-view phase-folded light curves. Unlike the baseline, which processes only the global flux sequence, this model jointly encodes the global and local views of each light curve along with their corresponding centroid-position features. This architecture is shown in Figure 3.

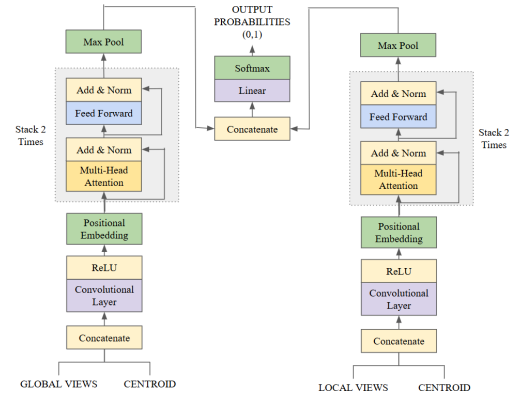


Figure 3. Transformer architecture

4.3.1 Multi-View Input Representation. Each training sample provides:

$$X = \{(f_g, c_g), (f_l, c_l)\}$$

where $f \in \mathbb{R}^T$ is normalized flux and $c \in \mathbb{R}^T$ is the centroid sequence identifying likely transit locations. The pairs are treated as two-channel time-series, allowing the network to jointly reason over flux shape and position.

4.3.2 ConvEmbedding for Local Context. Instead of embedding tokens independently, each view first passes through a small convolutional feature extractor (kernel size 5) to capture short-range structure prior to attention:

$$(f, c) \in \mathbb{R}^{2 \times T} \xrightarrow{\text{ConvEmbedding}} H \in \mathbb{R}^{T \times d_{\text{model}}}$$

This improves stability in noisy TESS signals and reduces reliance on attention to learn local patterns from scratch.

4.3.3 Positional Encoding and Self-Attention. We apply sinusoidal positional encodings to preserve time ordering, followed by two layers of Transformer Encoder blocks with multi-head self attention (8 heads, $d_{\text{model}} = 128$, feed-forward size 256). A pre-LayerNorm stabilization was added to prevent exploding attention activations observed in early experiments.

For each view, encoded representations are globally pooled via

$$h = \max_t(H_t)$$

which emphasizes the most discriminative temporal features (e.g. deepest transit region).

4.3.4 Multi-View Fusion and Classification. The model concatenates pooled local and global embeddings.

$$z = [h_{local}; h_{global}] \in \mathbb{R}^{2d_{model}}$$

This vector is fed into a two-layer MLP classifier to produce logits for two output classes.

4.3.5 Training. The model is trained using cross-entropy loss with Adam optimization (learning rate 10^{-5}), mini-batch size 32, and a stratified 75/15/15 train-val-test split. To improve generalization, we enable stochastic augmentation on the training subset only, including random mirroring, Gaussian noise, and circular shifts of the flux and centroid channels.

Model parameters are clipped (max-norm 1.0) each step to ensure training stability under attention, and early versions were monitored for non-finite activations to prevent divergence. Validation performance is tracked every epoch using accuracy and PR-AUC, the latter being critical for evaluating detection capability on rare planet cases.

4.4 Pseudo-Labeling for Semi-Supervised Training

To further exploit the large collection of unlabeled TESS Objects of Interest (TOIs), we incorporate a semi-supervised learning strategy based on **pseudo-labeling**. This method expands the effective training set using high-confidence predictions from an existing model.

After training the supervised transformer described above, we deploy it on the unlabeled TCE dataset and obtain prediction scores for each sample. Rather than using all predictions, we select only those whose confidence exceeds a strict threshold:

$$\hat{y} = \arg \max p(y|X), \quad \text{if } \max p(y|X) > \tau$$

where we use $\tau = 0.90$ for planet classification and $\tau = 0.99$ for non-planet classification to limit noise, avoid amplifying incorrect labels, and manage the rare event occurrences. The pseudo-labeled data is downsampled to a 3:1 ratio of Non-Planets:Planets, and concatenated with the original training dataset to create a new, combined dataset.

The transformer is then retrained on this combined dataset under the same supervised objective:

$$\mathcal{L}_{CE}(X, \hat{y})$$

while continuing to use stratified batching to preserve class imbalance. The classifier head is fully fine-tuned, and the encoder remains trainable, allowing the attention weights to adjust to the expanded data distribution.

4.4.1 Benefits. As Transformers generally improve with more data, this approach encourages the model to generalize beyond the limited number of confirmed planets and reduces overfitting to noisy labeled subsets. As pseudo-labels are derived from true TESS distributions, the model learns a broader representation of transit morphology, especially when it comes to improving decision boundaries for ambiguous cases where false positives resemble shallow planetary dips.

4.4.2 Risks and Mitigation. Psuedo-labeling can reinforce errors if low-confidence predictions are included. The confidence threshold and continued monitoring of validation metrics help avoid collapse and ensure that newly introduced samples improve performance rather than degrade it.

4.5 Self-Supervised Reconstruction + Transformer

While the supervised transformer leverages only labeled signals, the scarcity of confirmed planets (relative to false positives and unknown TCEs) motivates incorporating unlabeled light curves. We developed a self-supervised dual-view transformer model that learns transit related structure through a masked reconstruction objective inspired by BERT-style masked language modeling. This architecture is shown in Figure 4.

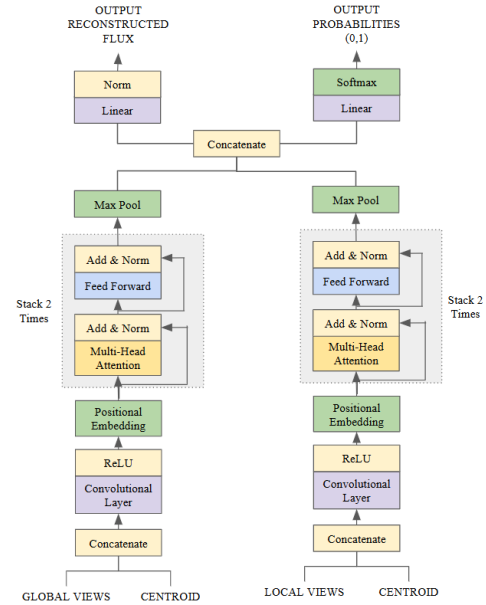


Figure 4. Self-supervised learning transformer architecture

4.5.1 Masked Dual-View Input. Unlabeled TESS samples provide the same two-view representation.

$$\{(f_g, c_g), (f_l, c_l)\}$$

where f denotes normalized flux and c denotes centroid signals. During pre-training, a random subset of flux values

(15% probability) is masked to zero:

$$\tilde{f}(t) = \begin{cases} 0, & t \in M \\ f(t) & \text{otherwise} \end{cases}$$

Centroid features remain unmasked and act as structural guidance about transit location.

The model learns to reconstruct the masked elements by minimizing masked-MSE loss.

$$\mathcal{L}_{mask} = \frac{\sum_{t \in M} (\hat{f}(t) - f(t))^2}{|M| + \epsilon}$$

4.5.2 DeepResidualEmbedding for Local Feature Extraction. Before attention, each view passes through a CNN-style residual embedding module that mimics the inductive bias proven effective in the baseline:

$$(2 \times T) \rightarrow Conv \rightarrow ResBlock \rightarrow H \in \mathbb{R}^{T \times d_{model}}$$

This stabilizes the reconstruction objective by encoding short-range flux-dips and smoothing instrumental noise.

4.5.3 Dual Transformer Encoders for Local and Global Streams. As with the supervised transformer, local and global views are processed through independent self-attention encoders (2 layers, 4 heads, $d_{model} = 128$). A sinusoidal positional encoding provides ordering information. Unlike the classifier transformer, no pooling is applied. The network outputs a sequence representation so that per-timestep reconstructions can be compared against masked ground truth.

4.5.4 Self-Supervised Objective and Representation Transfer. After pretraining for 20 epochs on the unlabeled dataset:

- The Transformer encoder is retrained.
- Reconstruction heads are discarded.
- A new classification head is attached to the pooled dual-view embeddings (max pooling along time).
- The encoder is initially frozen, enabling rapid convergence with limited labeled data.
- Fine-tuning proceeds for 10 epochs using cross-entropy loss on the known planet/false-positive labels.

This two-stage learning scheme allows the model to first acquire a generic representation of transit morphology using unlabeled TESS signals and then specialize on rare supervised examples.

4.5.5 Training Configuration. Pretraining uses an AdamW optimizer with a learning rate of 1×10^{-4} , batch size 64, and masked-MSE loss for both views. Fine-tuning uses AdamW with a reduced learning rate of 1×10^{-5} , batch size 32, and early-epoch encoder freezing to improve label efficiency. Evaluation follows the same stratified train/validation/test protocol as in other models, reporting accuracy and PR-AUC. PR-AUC offers the most meaningful metric for detecting rare planets.

5 Empirical Analysis

5.1 Research Questions

- RQ1: Can Transformers outperform CNNs in detecting exoplanets from TESS light curves?
- RQ2: Does self-attention and inclusion of global and local views improve recall for exoplanet classification?
- RQ3: How does data augmentation affect model performance and stability?
- RQ4: Can Self Supervised Learning (SSL) on unlabeled light curve data improve model performance?

5.2 Data Description

We use light curves from the Transiting Exoplanet Survey Satellite (TESS) mission, accessible through the Mikulski Archive for Space Telescopes (MAST). The dataset includes confirmed and candidate planetary transits along with stellar metadata such as luminosity and temperature.

We adopt a TESS dataset constructed by the ExoMiner team at NASA. It includes 280,000 transit crossing events from stellar sectors 1-67, spanning years 1-4. Each TCE is labeled, examples of labels include UNK: Unknown, CP: Confirmed Planet, KP: Known Planet, EB: Eclipsing Binaries, NEB: Nearby Eclipsing Binary, NTP: Nearby Eclipsing Binary. 72% TCE's (about 150,000) are unknown. Among the known TCE's, 69% of the TCE's are non-exoplanets, Figure 5 plots the binary class distribution.

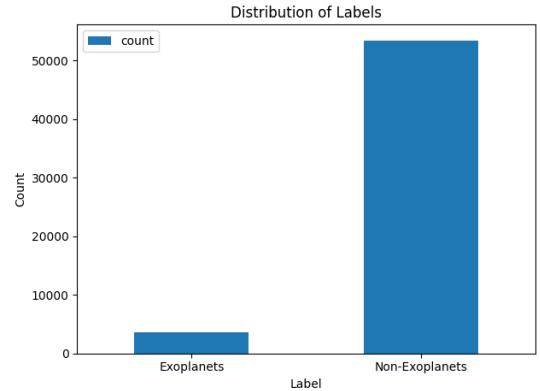


Figure 5. Distribution of Binary Classes

Since our classes are heavily imbalanced, and a vast majority of the TCE light curves are unknown, we will use supervised learning to train models on the known Exoplanets and Non-Exoplanets, and use self-supervised learning to pseudo-label the unknown light curves.

We plan to expand beyond the sample shard of the full dataset for final evaluation.

5.3 Evaluation Metrics

Due to the heavy class imbalance in the TESS dataset, we evaluated models and approaches with the following multiple classification metrics:

- Precision - $(\frac{TP}{TP+FP})$
- Recall - $(\frac{TP}{TP+FN})$
- Accuracy - $(\frac{TP+TN}{TP+TN+FP+FN})$
- PR-AUC (Area under Precision-Recall Curve)

5.4 Experiments

From the TESS dataset, we used stratified random sampling to create training, validation, and test datasets with a 70/15/15 split. The CNN and the Transformer with the multi-view inputs are trained with validation metrics reported after each epoch. Data augmentation is applied in the following proportions on the training set **only** for the Transformer w/ Data Augmentation: random mirroring (50%), Gaussian noise (50% with 0.01 standard deviation), and circular shifts of the flux and centroid channels (10%).

After the initial Transformer was trained, it was used to pseudo-label the TESS light curves which were labeled UNKNOWN (72% of the total TESS dataset) based on the different acceptance thresholds for the Planet (P) and Non-Planet (NP) classes. It was then re-trained on the combined dataset of the original training data and the down-sampled pseudo-labels (**18051** Potential NPs and **6017** Potential Ps).

All 4 models were evaluated on the test set on the classification metrics. (*The Transformer trained for reconstructing masked light curves was not included due to poor performance and time constraints*).

5.5 Experimental Results

We evaluated the performance of the CNN baseline and three transformer-based models (without augmentation, with data augmentation, and with SSL). Table 1 shows the experimental results for these models for the relevant metrics. Recall and AUC-PR are emphasized over accuracy and precision due to the imbalanced dataset and relative low risk of identifying false positives (1.8% confirmed planets in TESS dataset).

Table 1. Model Performance Comparison

Model	Accuracy	Precision	Recall	PR-AUC
CNN	0.97	0.78	0.77	0.83
Transformer	0.97	0.70	0.85	0.82
Transformer w/ Data Augmentation	0.97	0.72	0.82	0.83
Transformer (SSL)	0.96	0.71	0.86	0.81

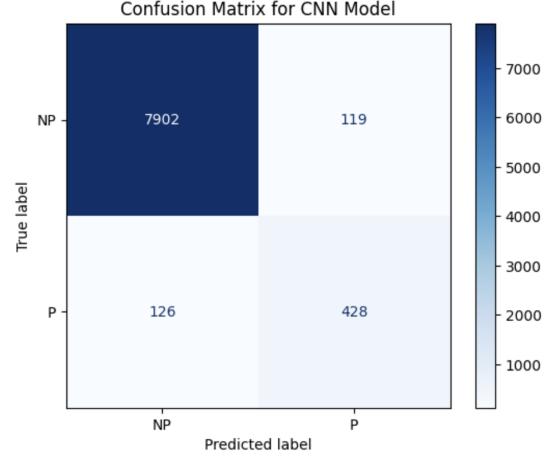


Figure 6. CNN achieves strong performance on the non-planet class and strong, but lower performance on the planet class

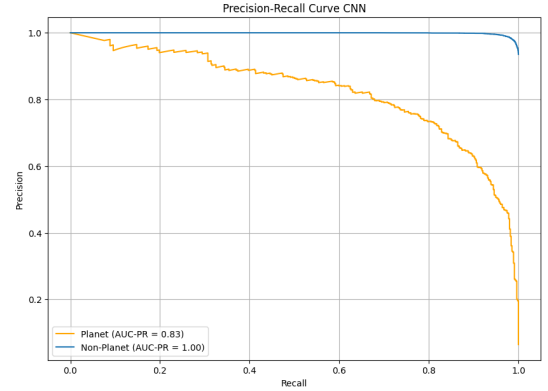


Figure 7. Precision-Recall curve for CNN baseline. The CNN achieves strong classification ability for the P class, given the imbalanced dataset.

Figure 6 demonstrates the CNN’s performance. Over 8021 instances, it achieves a recall of 0.77 and a precision 0.97.

Figure 7 indicates strong classification ability for the CNN, given the high PR-AUC of 0.83 for the P class despite the class imbalance.

Figure 8 shows the vanilla transformer’s performance is comparable to the CNN in terms of PR-AUC (0.82 vs 0.83). However, the transformer’s recall is much higher (0.85 vs 0.77)

Figure 9 shows the SSL-enhanced transformer’s performance, where it achieves the highest recall (0.86) among all other models, consistent with its ability to learn generalized transit patterns from unlabeled data. This improvement is accompanied by a decrease in precision and PR-AUC.

Across all experiments, the CNN has the highest precision. The CNN and the Transformer w/ Data Augmentation have

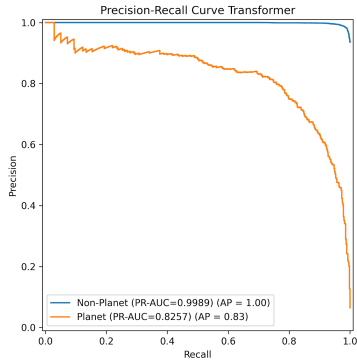


Figure 8. Precision-Recall curve for the vanilla transformer model. Even though the AUC-PR is similar to the CNNs, the transformer achieves a higher recall.

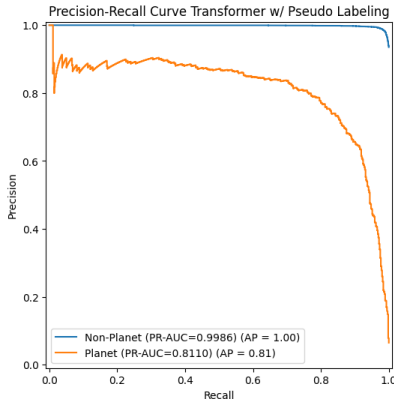


Figure 9. Precision-Recall curve for SSL-enhanced transformer. It achieved slightly lower AUC-PR compared to the CNN and transformer, but had the highest recall.

the highest PR-AUC (while only by 1-2%). However, the Transformers have considerably higher recall than the CNN - the Self Supervised Transformer has the highest at 0.86.

The results demonstrate that transformers show promise for large-scale exoplanet candidate detection, especially including the local view and the global view as inputs, with a tradeoff for precision. However, the transformers are more compute and time intensive to train in comparison to the CNN. The marginal results for metrics beside recall can inform approaches to detecting exoplanets based on different resource constraints. **RQ1, RQ2.**

Data augmentation can improve performance, balancing precision and recall for generalized exoplanet detection. However, further work can be done to evaluate the efficacy of different data augmentation strategies, and to ensure that data augmentation aligns with physical laws. **RQ3.**

Self-supervised learning with pseudo-labeling can slightly improve recall and is a worthwhile strategy for utilizing the

vast amount of unlabeled astronomical data and improving model performance on unseen data to identify exoplanets. Self-supervised learning with masking reconstruction did not perform well (0.17 PR-AUC), and this approach could be further explored with modifications to the masking strategy and loss optimization. **RQ4.**

A larger dataset would be necessary to make more informed, statistically significant conclusions about model performance. More labeled data or adjustments to the transformer architecture could enable them to surpass CNNs in tasks where precision takes priority.

6 Conclusions

6.1 Summary of Accomplishments

After gaining access to ~90GB of flux curve data from the TESS NASA mission, we were able to train two model architectures to identify exoplanets within this dataset. The first model, serving as the baseline, was the CNN which displayed an impressive AUC-PR of 0.83. Transformers were the primary focus of our method, with three variants evaluated: a baseline transformer, a transformer with data augmentation to address class imbalance, and a transformer augmented with self-supervised learning. With AUC-PRs of 0.82, 0.83, and 0.81, respectively, all three variants performed similarly to the CNN.

6.2 Extensions of Project

Early in the project, we considered using sparse attention variants like BigBird or Longformer to improve the efficiency of classification. After accessing the TESS data, it became clear that the N^2 cost of traditional self-attention would not scale beyond the means of our compute because the temporal length of data was less than presumed [1, 12]. Utilizing sparse attention was no longer considered until the light curves themselves were understood further.

The local and global views of the flux curves are generated by phase folding long series of flux data on top of itself to generate an aforementioned *average* representation of the curve. This means that without this data pre-processing step, the data would be very long temporal series of flux data, which may warrant the use of a sparse attention mechanism; therefore, a natural extension is to omit phase-folding and utilize a sparse-attention embedding of the unfolded light curve alongside the local and global embeddings.

6.3 Learning Takeaways

None of us had worked with astronomical data or exoplanets before, so becoming familiar with the space and the idea of TCEs was new for us. Additionally, we hadn't dealt with a class imbalance within a dataset quite like this before, so we learned new approaches like data augmentation and self-supervised learning with parallel tasks to take advantage of unlabeled data and improve performance.

7 Division of Work

- Gannon Smith: Transformer implementation, data augmentation, and project direction
- Alex Bartolozzi: Dataset management and processing, experimentation design, implementing transformer
- Nathan Zekarias: Implemented CNN baseline, abstract, and a portion of the results section
- Saipranav Avula: Data preprocessing, experimentation design, Self-Supervised Learning

All team members contributed equally to class reports and presentations.

8 Disclosure of GenAI Use

- ChatGPT and Claude were used in the debugging process of the Transformer and CNN notebooks to identify and diagnose problems.
- ChatGPT was used to help construct the README.md file for the Github Repository.
- ChatGPT and Consensus were used to assist in the search and interpretation of the cited papers within the report. All writing is our own.

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